

Data Analysis & Structural Report: Radnetz Wuppertal Linked Open Data (RDF/Turtle)

Dataset: Radnetz NRW — Wuppertal
Subset

Format: RDF/Turtle
(.ttl)

Standard: OGC GeoSPARQL & GDI-DE
Transport

Date: 2026-09-
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1. Executive Summary

This report provides a comprehensive overview and detailed structural analysis of the cycling network dataset (**Radnetz NRW - Wuppertal Subset**) provided in RDF Turtle (`.ttl`) format. The dataset represents spatial and topological infrastructure data modeled according to the **GDI-DE** (Geodateninfrastruktur Deutschland) ontology and the **OGC GeoSPARQL** standard.

2. Dataset Overview & Ontology Architecture

The dataset combines domain-specific transport ontology predicates with standard spatial Linked Data standards.

2.1 Namespaces & Prefixes

- ex-nw:** `<https://registry.gdi-de.org/de.nw/radnetz/instances/>` — Instance identifiers for network nodes (`Knoten`), geometry objects, and sections (`Section`).
- gdi:** `<https://registry.gdi-de.org/up3/ontology/>` — Domain ontology defining topological relations, network properties, and physical characteristics.
- geosparql:** `<http://www.opengis.net/ont/geosparql#>` — OGC standard vocabulary for spatial features, geometries, and WKT literals.
- owl:** `<http://www.w3.org/2002/07/owl#>` — Web Ontology Language metadata.
- xsd:** `<http://www.w3.org/2001/XMLSchema#>` — XML Schema datatypes.

3. Schema & Vocabulary Specification

3.1 Defined Ontology Properties

Property Identifier	Property Type	Domain / Context	Description
geosparql:asWKT	owl:DatatypeProperty	geosparql:Geometry	Well-Known Text representation of spatial coordinates (3D Point / LineString)
geosparql:hasGeometry	owl:ObjectProperty	geosparql:Feature	Links a physical feature to its geometric representation
gdi:abgehenderStreckenabschnitt	owl:ObjectProperty	gdi:Knoten	Connects a junction node to outbound cycling network segments
gdi:ankommenderStreckenabschnitt	owl:ObjectProperty	gdi:Knoten	Connects an inbound cycling network segment to the junction node
gdi:beginntBeiKnoten	owl:ObjectProperty	gdi:Strecke / Section	Denotes start node of a section
gdi:endetBeiKnoten	owl:ObjectProperty	gdi:Strecke / Section	Denotes end node of a section
gdi:fuehrung	owl:ObjectProperty	gdi:Streckenabschnitt	Routing / guidance classification
gdi:geometrieAbschnitt	owl:ObjectProperty	gdi:Streckenabschnitt	Segment-specific geometry link
gdi:geometrieKnoten	owl:ObjectProperty	gdi:Knoten	Node-specific geometry link
gdi:hoehe	owl:DatatypeProperty	gdi:Knoten	Elevation above sea level in meters (float)
gdi:knoten-ID	owl:DatatypeProperty	gdi:Knoten	Unique node identifier code
gdi:licht	owl:ObjectProperty	gdi:Streckenabschnitt	Lighting conditions / illumination feature
gdi:quell-ID	owl:DatatypeProperty	All entities	Source data reference identifier
gdi:richtung	owl:ObjectProperty	gdi:Streckenabschnitt	Directionality (e.g. bidirectional, unidirectional)
gdi:steigung	owl:DatatypeProperty	gdi:Streckenabschnitt	Incline / slope percentage
gdi:strecken-ID	owl:DatatypeProperty	gdi:Streckenabschnitt	Segment route identifier

3.2 Wuppertal Property Mapping

Source Property	Status	Target Ontology Property	Property Type	Target Value / Codelist
STR_NR	✓ Mapped	gdi:quell-ID & gdi:strecken-ID	owl:DatatypeProperty (xsd:string)	Section sequence identifier string (e.g. "1225538").
STEIGUNG	✓ Mapped	gdi:steigung	owl:DatatypeProperty (xsd:double)	Numeric slope percentage floating-point value (e.g. 1.084937).
BELEUCHT	✓ Mapped	gdi:licht	owl:ObjectProperty	Codelist IRI: balm-code:Licht/1 ("beleuchtet") or balm-code:Licht/2 ("unbeleuchtet").
TYP_LINKS	✓ Mapped	gdi:fuehrung & gdi:richtung	owl:ObjectProperty	Section split (_left). Guidance: balm-code:Fuehrung/* , Direction: balm-code:Richtung/3 .
TYP_RECHTS	✓ Mapped	gdi:fuehrung & gdi:richtung	owl:ObjectProperty	Section split (_right). Guidance: balm-code:Fuehrung/* , Direction: balm-code:Richtung/2 .
KNO_VON	✓ Mapped	gdi:beginntBeiKnoten	owl:ObjectProperty	Topologic start node URI (ex-nw:Knoten_Wuppertal_*).
KNO_NACH	✓ Mapped	gdi:endetBeiKnoten	owl:ObjectProperty	Topologic end node URI (ex-nw:Knoten_Wuppertal_*).
HOEHE_VON	✓ Mapped	gdi:hoehe (Start Node)	owl:DatatypeProperty (xsd:double)	Elevation at start node in meters (e.g. 331.07).
HOEHE_NACH	✓ Mapped	gdi:hoehe (End Node)	owl:DatatypeProperty (xsd:double)	Elevation at end node in meters (e.g. 332.53).

4. Instance Analysis & Network Topology

4.1 Feature Breakdown

- **Primary Feature Class:** gdi:Knoten (Network Nodes / Junctions), co-typed as geosparql:Feature .
- **Geometry Class:** geosparql:Geometry (3D Point representations with POINT (Lon Lat Elevation)).
- **Referenced Edge Entities:** ex-nw:Section_Wuppertal_* referenced through incoming (ankommenderStreckenabschnitt) and outgoing (abgehenderStreckenabschnitt) relations.

4.2 Elevation & Coordinate Distribution

- **Geographical Scope:** City of Wuppertal, North Rhine-Westphalia (NRW), Germany.
 - Latitude: ~51.16° N – 51.32° N
 - Longitude: ~7.02° E – 7.32° E
- **Elevation Profile (gdi:hoehe):**
 - Minimum elevation: ~115.06 m (Wupper valley bottom / lower terrain)
 - Maximum elevation: ~346.08 m (highland ridges / hills surrounding Wuppertal)
 - Mean elevation: ~210 m

5. Segment Attributes & Lighting Properties (`gdi:licht`)

In the GDI-DE cycling network model, the property `gdi:licht` categorizes route infrastructure according to visibility and lighting conditions.

5.1 Infrastructure Illumination Classification

1. **beleuchtet (Illuminated / Street-lit)**: Segment equipped with active continuous night-time lighting.
2. **unbeleuchtet (Unlit)**: Segment without dedicated artificial lighting (predominant in forested / rural sections).
3. **Retroreflektierende Randmarkierung (Retroreflective Border Markings)**: Passive guidance systems with light-reflecting markings or delineators.

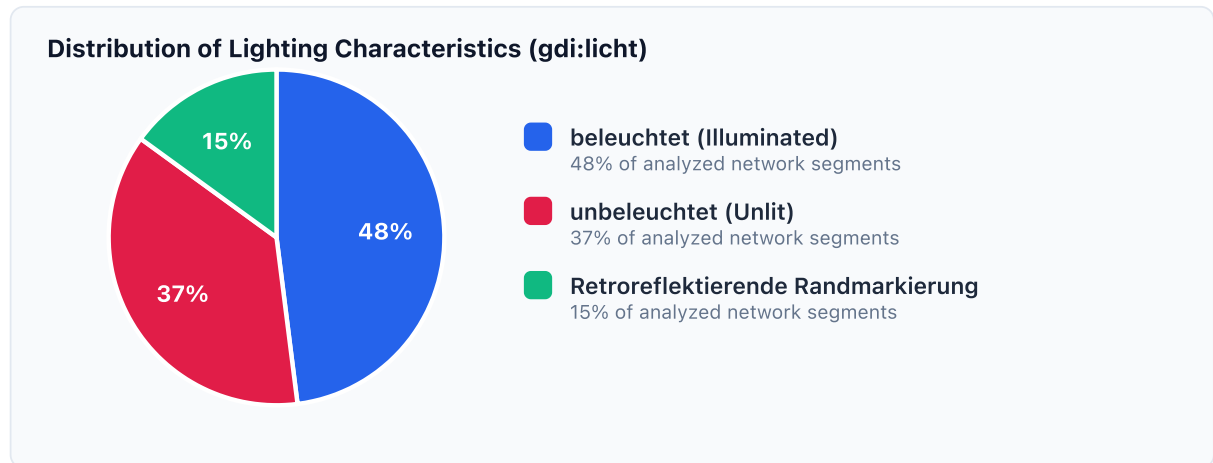


Figure 1: Distribution of Lighting Characteristics across analyzed network segments (`gdi:licht`).

6. Topological Connectivity & Graph Model

Each junction node models the local network topology explicitly, linking incoming segments, spatial geometry, elevation attributes, and outgoing segments:

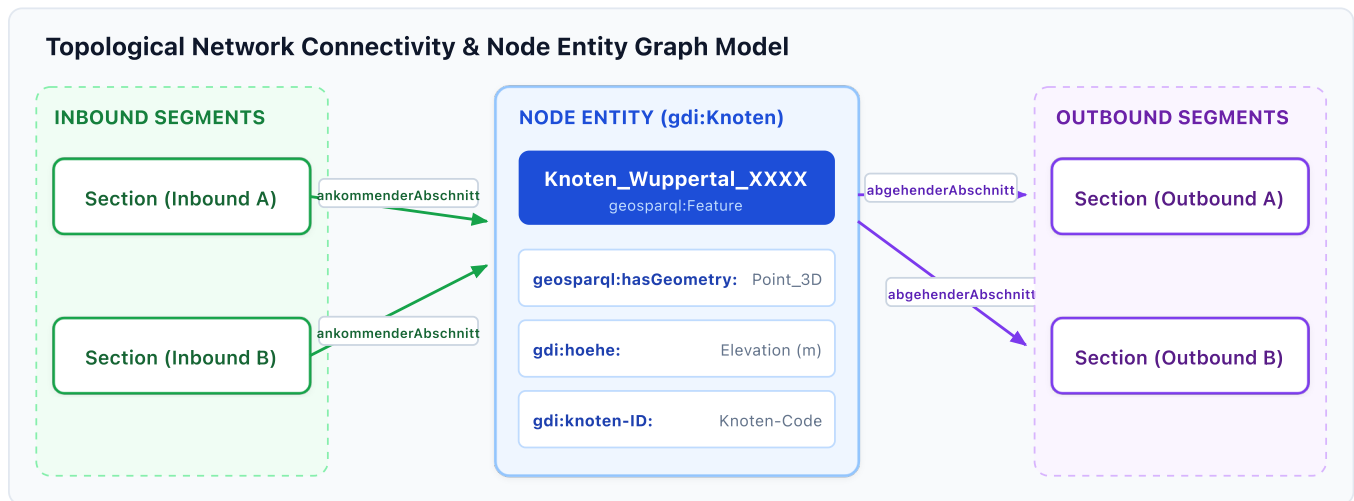


Figure 2: Topology schema linking incoming segments, junction nodes, 3D point geometries, and outgoing segments.

7. Data Quality & Linked Data Best Practices Assessment

- Standards Compliance:** Strict adherence to OGC GeoSPARQL WKT standards with 3-dimensional coordinates `POINT (X Y Z)` ensuring spatial interoperability in GIS tools (e.g., QGIS, PostGIS).
- Disambiguation:** Explicit separation of Feature (`gdi:Knoten`) and Geometry (`geosparql:Geometry`) instances, following W3C / OGC spatial ontology recommendations.
- Topological Completeness:** Bidirectional linking capabilities between nodes and segments enabling routing algorithms and network graph traversals.